

The area of a triangle, and trigonometry in three dimensions

Cambridge insert: one area formula for every triangle, applied inside solids.

LESSON 29–30

2 × 40 MIN

GEOMETRY 10, §13 (EXTENSION)

IGCSE E6.5 · E6.6

LESSON CLOCK

10'

16'

26'

24'

20'

4

The area formula	10 min
Heron	16 min
Sections of solids	26 min
Mixed 3-D problems	24 min
Practice	20 min
Homework	4 min

LEARNING OBJECTIVES

- Use $S = \frac{1}{2}ab \sin C$ for any triangle.
- Use Heron's formula when three sides are known.
- Find the area of a triangular section of a solid.
- Combine the area formula with the sine and cosine rules in a 3-D problem.

TEXTBOOK

National	pp. 137–142
Cambridge	Core & Extended, pp. 300–312
Workbook	IGCSE Exercise 6.5

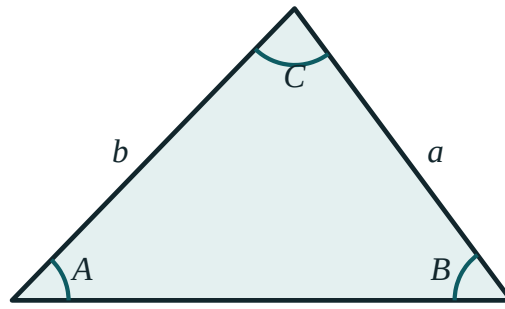
01

Explanation

One formula for every triangle

$$S = \frac{1}{2}ab \cdot \sin C$$

Two sides and the angle **between them**. It reduces to $\frac{1}{2}bh$ when $C = 90^\circ$, because $\sin 90^\circ = 1$ — the familiar formula is the special case.



$$a / \sin A = b / \sin B = c / \sin C$$

$$a^2 = b^2 + c^2 - 2bc \cdot \cos A$$

The labelling convention: side a opposite angle A. The area uses the angle between the two named sides.

THE ANGLE MUST BE THE INCLUDED ONE

$\frac{1}{2}ab \cdot \sin A$ is wrong: A is not between a and b. Name the two sides first, then take the angle they enclose.

Heron's formula

When all three sides are known and no angle is:

$$S = \sqrt{s(s-a)(s-b)(s-c)}, \text{ where } s = \frac{a+b+c}{2}$$

For 5, 6, 7: $s = 9$ and $S = \sqrt{9 \cdot 4 \cdot 3 \cdot 2} = \sqrt{216} \approx 14.7$.

WHICH TO USE

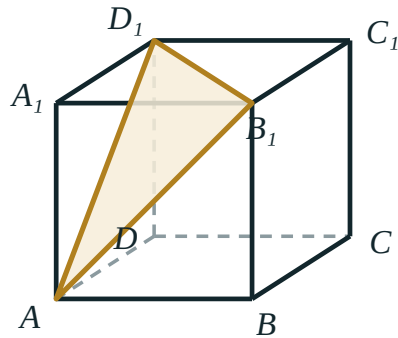
Two sides and the included angle → the sine formula. Three sides → Heron, or find an angle by the cosine rule and then use the sine formula. Both give the same number.

Areas of sections

The sections of Lesson 12–13 are polygons in space; their areas are computed by splitting into triangles and applying the formula to each.

Example. Cube of side 6. The section through A, C, B_1 is an equilateral triangle of side $6\sqrt{2}$. Using the formula with $C = 60^\circ$:

$$S = \frac{1}{2} \cdot 72 \cdot \sin 60^\circ = 36 \cdot \frac{\sqrt{3}}{2} = 18\sqrt{3} \approx 31.2$$



the section
A B₁ D₁ is
an equilateral
triangle

The triangular section, and the triangle drawn flat beside it.

Mixed problems

A full 3-D problem chains the three tools:

1. Find the lengths of the section's sides by Pythagoras in the solid.
2. Find an angle by the cosine rule in the extracted triangle.
3. Find the area by $\frac{1}{2}ab \sin C$.

Example. Cuboid $4 \times 6 \times 12$. Section through $A(0,0,0)$, $B(4,0,0)$ and $C_1(4,6,12)$: $AB = 4$, $BC_1 = \sqrt{36 + 144} \approx 13.42$, and $AB \perp BC_1$ because AB is perpendicular to the whole face BCC_1B_1 . So the area is $\frac{1}{2} \cdot 4 \cdot 13.42 \approx 26.8$.

02 Worked examples

EXAMPLE 1 Find the area of a triangle with $a = 7$, $b = 9$, $C = 40^\circ$.

$$① \quad S = \frac{1}{2} \cdot 7 \cdot 9 \cdot \sin 40^\circ$$

$$② \quad = 31.5 \times 0.6428$$

$$③ \quad \approx 20.2$$

ANSWER

≈ 20.2 square units

EXAMPLE 2 Find the area of a triangle with sides 8, 11 and 13.

$$① \quad s = 16$$

$$② \quad S = \sqrt{16 \cdot 8 \cdot 5 \cdot 3}$$

$$③ \quad = \sqrt{1920} \approx 43.8$$

ANSWER

≈ 43.8

EXAMPLE 3 Cube of side 6. Find the area of the section through A, C, B_1 .

- ① All three sides are face diagonals, $6\sqrt{2}$.
Equilateral.

② $S = \frac{1}{2} \cdot 72 \cdot \sin 60^\circ$

③ $= 18\sqrt{3} \approx 31.2$

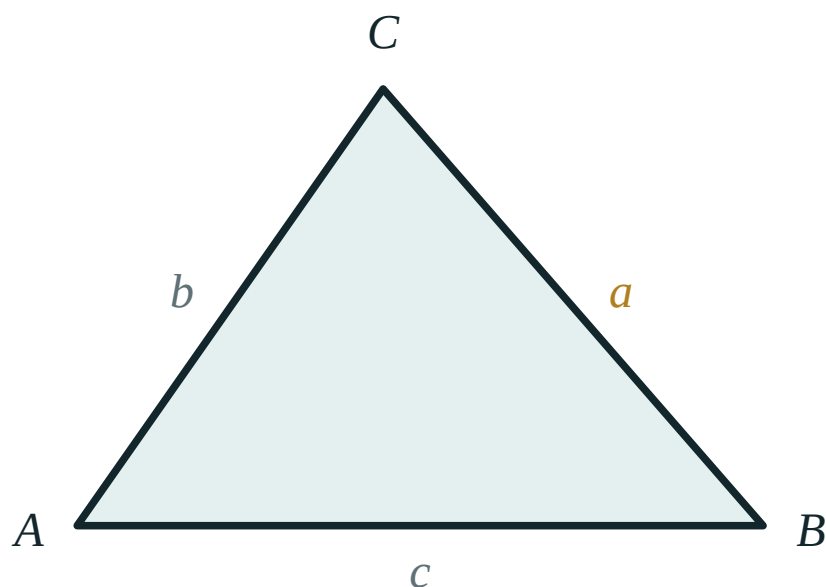
ANSWER

$$18\sqrt{3} \approx 31.2$$

03 **Interactive model**

Extract the triangle from the solid and draw it flat before any calculation.

Two sides and the included angle



b 7

c 9

A 55°

a 7.6

B 49°

C 76°

area 25.8

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Area of a triangle	Uchburchak yuzasi	Площадь треугольника
Included angle	Ichki burchak	Угол между сторонами
Heron's formula	Geron formulasi	Формула Герона
Semi-perimeter	Yarim perimetr	Полупериметр
Triangular section	Uchburchakli kesim	Треугольное сечение
Sine rule	Sinuslar teoremasi	Теорема синусов
Cosine rule	Kosinuslar teoremasi	Теорема косинусов
Three-dimensional problem	Fazoviy masala	Пространственная задача
Extracted triangle	Ajratilgan uchburchak	Выделенный треугольник

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The area formula $\frac{1}{2} ab \sin C$ needs:

any angle

the angle between a and b

the largest angle

a right angle

Heron's formula needs:

two sides and an angle

three sides

two angles

the height

s in Heron's formula is:

the perimeter

half the perimeter

the longest side

the area

Area of a triangle with $a = 6$, $b = 8$, $C = 30^\circ$:

The section of a cube through A , C , B_1 is:

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Area with $a = 4$, $b = 5$, $C = 90^\circ$

ANSWER 10

2. Area with $a = 6$, $b = 8$, $C = 30^\circ$

ANSWER 12

3. Area with $a = 10$, $b = 10$, $C = 60^\circ$

ANSWER $25\sqrt{3}$

4. Heron: sides 3, 4, 5

ANSWER 6

5. Heron: sides 5, 6, 7

ANSWER $\sqrt{216} \approx 14.7$

6. s for sides 8, 11, 13

ANSWER 16

7. Area with $a = 12$, $b = 5$, $C = 90^\circ$

ANSWER 30

Medium · the standard question

7 PROBLEMS

1. Area with $a = 7$, $b = 9$, $C = 40^\circ$

ANSWER ≈ 20.2

2. Heron: sides 8, 11, 13

ANSWER ≈ 43.8

3. Cube side 6: area of section ACB_1

ANSWER $18\sqrt{3}$

4. Cube side 4: area of the diagonal section ABC_1D_1

ANSWER $16\sqrt{2}$

5. Triangle 9, 12, 15: area

ANSWER 54

6. Two sides 10 and 14 enclose 120° . Area

ANSWER $35\sqrt{3} \approx 60.6$

7. Cuboid $4 \times 6 \times 12$: area of section ABC_1

ANSWER ≈ 26.8

Hard · stretch and reason

7 PROBLEMS

1. Cube side a : area of the regular hexagonal section

ANSWER $\frac{3\sqrt{3}}{4}a^2$

2. Square pyramid, base 10, height 12: area of one lateral face

ANSWER 65

3. Same pyramid: total lateral surface area

ANSWER 260

4. Triangle with sides 13, 14, 15: area and the height on the 14-side

ANSWER 84, height 12

5. Find the angle when $a = 8$, $b = 10$ and the area is 20

ANSWER 30° or 150°

6. Regular tetrahedron edge 6: total surface area

ANSWER $36\sqrt{3}$

7. Prove $S = \frac{1}{2}ab \sin C$ from $\frac{1}{2}bh$

ANSWER $h = a \sin C$

07 Homework

Homework — 5 tasks

Draw the extracted triangle for every 3-D question.

- Find the area of a triangle with $a = 11$, $b = 6$, $C = 55^\circ$.
- Find the area of a triangle with sides 7, 9 and 12 by Heron's formula.
- Cube of side 8: find the area of the section through A , C , B_1 .
- Square pyramid, base 12, height 8: find the area of one lateral face and the total lateral area.

5. Two sides 9 and 12 enclose an angle whose sine is 0.6. Find the area and both possible angles.